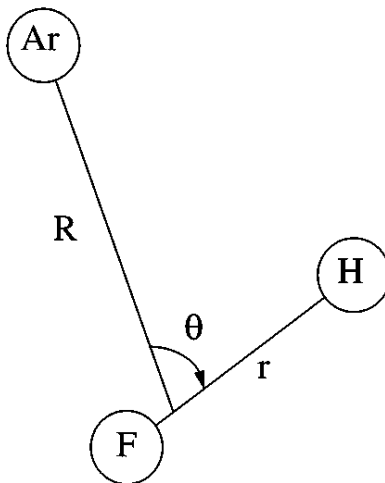


Jacobi Coordinates for Ar–HF

Ar–HF: we need $3N - 6 = 3 \cdot 3 - 6 = 3$ coordinates to describe its geometry.



$$\mathbf{r} = \mathbf{r}_F - \mathbf{r}_H \quad (1)$$

$$\mathbf{r}_{\text{CM}}^{\text{HF}} = \frac{m_H \mathbf{r}_H + m_F \mathbf{r}_F}{m_H + m_F} \quad (2)$$

$$\mathbf{R} = \mathbf{r}_{\text{Ar}} - \mathbf{r}_{\text{CM}}^{\text{HF}} \quad (3)$$

$$\theta = \arccos \frac{\mathbf{r} \cdot \mathbf{R}}{rR} \quad (4)$$

If we fix r , we are left with just two coordinates, (R, θ) .